

Package ‘BayGMST’

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Title: Bayesian Reconstruction of Global Mean Surface Temperature
Version: 0.1.0
Maintainer: Tyler Bagwell <teb6@rice.edu>
License: GPL (≥ 3)

READ THIS FIRST

This is a **hand-typeset preview**, not the real, authoritative CRAN reference manual. It was produced in a sandbox with **no R installation available**, so it could not be generated the normal way (`devtools::document()` to build `man/*.Rd`, then `devtools::build_manual()` / R CMD Rd2pdf to render this PDF from those `.Rd` files). Content below was written by hand to mirror that output as closely as possible, from the same source descriptions as the `roxygen2` comments in R/.

Before treating this as final: run the two commands above (see `CRAN-READINESS.md` at the package root for the full checklist) and let the real output supersede this file.

Text marked **[GAP]** like [this](#) is a genuine information gap I could not resolve myself. Text marked **[FLAG]** like [this](#) is a decision, judgment call, or unresolved issue from the original code that needs Tyler’s or Julien’s sign-off, not something I decided unilaterally.

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1 Statistical Model

[FLAG] This section is not part of a standard CRAN Rd2pdf reference manual – those only ever contain per-topic help pages, never narrative model exposition. It’s included here because it was explicitly requested, to connect the code to the underlying statistics. Consider moving it into `vignettes/baygmst-intro.Rmd` (or a separate paper/preprint) rather than keeping it in the reference manual once the real, roxygen2-generated version of this document is built, since R CMD Rd2pdf has no mechanism to reproduce a hand-added section like this one.

Global mean surface temperature (GMST) is a fundamental state variable of Earth’s climate system and a central target of paleoclimate reconstruction efforts. Because direct instrumental measurements extend back only about 150 years, estimates of GMST over longer periods must be inferred from indirect observations preserved in paleoclimate archives — tree rings, glacier ice, corals and sclerosponges, lake and marine sediments, and historical documents (*Mann et al., 1999; Moberg et al., 2005*). A variety of statistical strategies have been developed to extract the climate signal contained in these “proxies” (*Tingley et al., 2012; Smerdon and Pollack, 2016*), whether the target is a spatial climate field (*Smerdon et al., 2023*) or an aggregated index such as GMST.

BayGMST’s inference framework builds on the work of *Barboza et al. (2014, 2019)* and *Wang (2020)*, who utilized a reduced proxy Bayesian approach that leverages not only the climate proxy network, but also estimates of past forcings (e.g., greenhouse gas concentrations, solar, and volcanic activity), which are known to influence GMST.

Specifically, their Bayesian hierarchical modeling is based on computing a “reduced proxy” $\mathbf{P} = (P_1, \dots, P_t, \dots, P_n)^T$ which is an aggregate of the available paleoclimate proxy observations. Several methods exist to do so, like principal component regression, partial least squares, sliced inverse regression, or a simple composite (*Barboza et al., 2019*). This quantity is then modeled as autoregressive and linearly dependent on the GMST, denoted $\mathbf{T} = (T_1, \dots, T_t, \dots, T_n)^T$ (subscript t refers to time in years CE). This is the model’s data level:

$$P_t = \phi_P P_{t-1} + \alpha_0 + \alpha_1 T_t + \sigma_P \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, 1) \quad (1)$$

where ϕ_P denotes the lag-1 autocorrelation of P , the α ’s are coefficients, and σ_P controls the amplitude of the unit-variance noise term ϵ . The process level models T_t as autoregressive and linearly dependent on the climate forcings G_t (greenhouse gas concentrations), V_t (related to volcanic aerosol optical depth), and S_t (solar irradiance) – already transformed, see `transform_forcings()`:

$$T_t = \phi_T T_{t-1} + \beta_0 + \beta_G G_t + \beta_V V_t + \beta_S S_t + \sigma_T \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} N(0, 1) \quad (2)$$

where ϕ_T denotes the lag-1 autocorrelation of $\mathbf{T}$, the β ’s are constant coefficients, and σ_T governs the amplitude of η_t . Gaussian white noise for both $\boldsymbol{\eta}$ and $\boldsymbol{\epsilon}$ provides good results (*Wang, 2020*). Here $|\phi_P|, |\phi_T| < 1$ and $\sigma_P, \sigma_T > 0$.

Mapping to the code

Notation above	Stan parameter	R argument / role
T_t	y (transformed parameter)	instrumental_T (observed part), y_mis (reconstructed part)
P_t	z (data)	proxy – see <code>reduce_proxies()</code>
ϕ_P	phi_R	—
σ_P	sigma_z	—
α_0, α_1	alpha0, alpha1	—
ϕ_T	phi_T	—
σ_T	sigma_y	—
$\beta_0, \beta_G, \beta_V, \beta_S$	beta0, betaG, betaV, betaS	—
G_t, V_t, S_t	G, V, S (data)	forcing_G, forcing_V, forcing_S

[FLAG] The Stan variable is named `phi_R/sigma_z` (“R” for “reduced proxy,” matching this specific implementation’s single-series design) rather than `phi_P/sigma_P` as used above for readability – same quantity,

different name, my own notational choice for this document, not a claim about how any cited paper names it. I did not rename the Stan code itself, to avoid touching model code beyond what was already agreed.

Priors

As specified in the Stan model’s `model` block:

$$\begin{aligned}\alpha_0, \alpha_1, \beta_0, \beta_G, \beta_V, \beta_S &\sim N(0, 0.5) \\ \phi_P \text{ (phi_R)}, \phi_T &\sim N(0, 0.3) \\ \sigma_P \text{ (sigma_z)}, \sigma_T \text{ (sigma_y)} &\sim \text{Exponential}(1)\end{aligned}$$

[FLAG] These weakly informative priors are exactly as inherited from the original `BayGMST_v1.0.stan`; I have not evaluated their sensitivity or appropriateness – that judgment belongs to Tyler/Julien.

Forcing data

Per [Barboza et al. \(2019\)](#) (§2.3), the three external forcings used in this family of models are:

- **Volcanic forcing** (V_t): zonal-mean stratospheric aerosol optical depth at 550 nm from the eVolv2k dataset ([Toohey and Sigl, 2017](#)), covering 500 BCE–1900 CE, area-weighted and averaged to a global annual estimate.
- **Solar forcing** (S_t): total solar irradiance from the SATIRE-H reconstruction ([Vieira, L. E. A. et al., 2011](#)), decadal before 1940 CE and daily thereafter, interpolated to annual resolution.
- **Greenhouse-gas forcing** (G_t): hemispheric-mean atmospheric CO₂ mole fraction (ppm) at annual resolution ([Meinshausen et al., 2017](#)), extended further back using ice-core-based compilations such as the PMIP4-CMIP6 `past1000` forcing set ([Jungclaus et al., 2017](#)).

[FLAG] This description is [Barboza et al. \(2019\)](#)’s own account of *their* forcing data, offered here as the best available explanation of where this package’s `data/forcing.csv` most likely originates, given the shared authorship and methodological lineage. I have not independently traced `data/forcing.csv`’s actual provenance (it carries no metadata/header documenting its source), so treat this as a well-informed inference, not a confirmed fact – Tyler/Julien can confirm directly.

[FLAG] [Barboza et al. \(2019\)](#) (§3.2) themselves use the transforms $\tilde{C}_t = \log(C_t)$ (greenhouse gases) and $\tilde{V}_t = \log(-V_t + 1)$ (volcanic), citing [Barboza et al. \(2014\)](#) for the choice. `transform_forcings()` instead implements $G' = \text{co2_coef} \times \log(G/\text{co2_c0})$ and $V' = -|\text{vol_coef}|(1 - e^{-V})$ (see `?transform_forcings`) – the same log-CO₂ spirit, but not an identical functional form, and a different saturating form for volcanic forcing. I have not read [Wang \(2020\)](#) or [Barboza et al. \(2014\)](#) directly, so I can’t say whether this divergence originates in one of those or is specific to Tyler’s own `BayGMST_v1.0.R`; worth checking directly.

Note on the instrumental period

[FLAG] In `baygmst.stan`, `y[idx_obs] = y_obs` sets the latent temperature state exactly equal to the raw instrumental values for observed years – i.e. instrumental temperature is treated as error-free, with no separate measurement-noise parameter. This is a modeling choice worth being aware of when interpreting instrumental-period fit statistics from `summary.baygmst_fit()`.

2 R topics documented:

BayGMST-package	BayGMST: Bayesian Reconstruction of Global Mean Surface Temperature
cv_baygmst	K-fold cross-validate the BayGMST model over the instrumental period
fit_baygmst	Fit the BayGMST Bayesian hierarchical AR(1) reconstruction model
plot_cv	Plot k-fold cross-validation reconstructions
plot_posterior_densities	Posterior density plots for a fitted BayGMST model
plot_reconstruction	Plot a BayGMST reconstruction
plot_trace	Trace plots for a fitted BayGMST model
print.baygmst_cv	Print method for "baygmst_cv" objects
print.baygmst_fit	Print a fitted BayGMST model
print.summary.baygmst_fit	Print method for "summary.baygmst_fit" objects
reconstruct	Extract a tidy reconstruction from a fitted BayGMST model
reduce_proxies	Reduce a multi-proxy network to a single representative proxy series
summary.baygmst_fit	Summarize a fitted BayGMST model
transform_forcings	Transform raw radiative forcings for use with <code>fit_baygmst()</code>

3 BayGMST-package

BayGMST: Bayesian Reconstruction of Global Mean Surface Temperature

Description

Fits a Bayesian hierarchical AR(1) state-space model relating global mean surface temperature (GMST) to a reduced paleoclimate proxy network, instrumental temperature observations, and radiative forcings (greenhouse gases, volcanic aerosols, and solar irradiance). Missing pre-instrumental temperatures are estimated jointly with the model parameters using Stan via the `instantiate` and `cmdstanr` packages.

Model fitting requires CmdStan

`fit_baygmst()` and `cv_baygmst()` compile and sample a Stan model, which requires a working CmdStan installation. Check with `instantiate::stan_cmdstan_exists()` first. `reduce_proxies()` and `transform_forcings()` do not require CmdStan.

Provenance

Restructured from a collection of top-level analysis scripts (see `NEWS.md` for exactly what changed, and `inst/legacy-scripts/` for the untouched originals). **[GAP]** The **Description:** field's exact public-facing wording in `DESCRIPTION` was drafted by me from the `README` and this package's own code – it has not been reviewed or approved by Tyler or Julien. Please read and edit it before any submission.

Author(s)

Tyler Bagwell <teb6@rice.edu> (maintainer), Julien Emile-Geay <julieneg@usc.edu>. **[FLAG]** These are the only two contributors visible in `git log`; add others if appropriate.

4 transform_forcings

Transform raw radiative forcings for use with `fit_baygmst()`

Description

Applies the forcing transformations used throughout the BayGMST pipeline: volcanic forcing is mapped to a negative saturating form, greenhouse-gas (CO₂) forcing is log-transformed relative to a reference concentration, and solar forcing is centered on its own mean. Ported from the normalization block near the top of the original `BayGMST_v1.0.R` script.

Usage

```
transform_forcings(G, V, S, co2_c0 = 280, co2_coef = 5.35, vol_coef = 25)
```

Arguments

G	Numeric vector of raw greenhouse-gas forcing (e.g. atmospheric CO ₂ concentration in ppm).
V	Numeric vector of raw volcanic forcing (e.g. aerosol optical depth). Same length as G .
S	Numeric vector of raw solar forcing (e.g. total solar irradiance). Same length as G .
co2_c0	Reference/baseline CO ₂ concentration used in the log transform. Default 280 (ppm).
co2_coef	Multiplier applied to the log CO ₂ ratio. Default 5.35.
vol_coef	Multiplier applied to the volcanic saturating transform. Default 25.

Details

$$V' = -|\text{vol_coef}| \times (1 - e^{-V}), \quad G' = \text{co2_coef} \times \log(G/\text{co2_c0}), \quad S' = S - \bar{S}$$

Value

A list with three numeric vectors, **G**, **V**, and **S** – the transformed forcings.

Examples

```
fc <- transform_forcings(
  G = c(280, 300, 340, 400),
  V = c(0, 0.05, 0.02, 0.10),
  S = c(1360, 1361, 1360.5, 1361.2)
)
fc$G
```

5 reduce_proxies

Reduce a multi-proxy network to a single representative proxy series

Description

Collapses a matrix of paleoclimate proxy records into a "representative proxy" series suitable for `fit_baygmst()`, following the segmented-composite approach of [Barboza et al. \(2019\)](#), ported from `utils/PAGES2k_reducedProxy_UNSC.R`. The proxy network is broken into overlapping time segments of `chunk` years; within each segment, available proxies are regressed against instrumental temperature during `calib_years` using the chosen `method`. The final composite series averages across all segments covering a given year.

Usage

```
reduce_proxies(proxy_matrix, years, temp_calib, calib_years,
method = c("PCR", "LASSO", "sPCR", "SPLS", "SIR"),
chunk = 250, target_adj_r2 = 0.7, max_na_frac = 0.05,
n_components = 3, spls_eta = seq(0.1, 0.9, 0.1))
```

Arguments

<code>proxy_matrix</code>	Numeric matrix/data.frame, one row per year (matching <code>years</code>), one column per proxy. No year column.
<code>years</code>	Integer vector of calendar years, one per row of <code>proxy_matrix</code> .
<code>temp_calib</code>	Numeric vector, instrumental temperature during <code>calib_years</code> .
<code>calib_years</code>	Integer vector, calibration years (subset of <code>years</code> ; <code>length(calib_years) == length(temp_calib)</code>).
<code>method</code>	One of "PCR" (default), "LASSO", "sPCR", "SPLS", or "SIR". "SIR" is experimental — see Details.
<code>chunk</code>	Segment length in years. Default 250. Every segment must fully contain <code>calib_years</code> , or <code>reduce_proxies()</code> errors (see Details).
<code>target_adj_r2</code>	For "PCR": adjusted- R^2 stopping threshold for adding principal components. Default 0.70.
<code>max_na_frac</code>	Proxies missing more than this fraction of a segment are dropped from it. Default 0.05.
<code>n_components</code>	For "sPCR" only: number of supervised components (scalar or per-segment vector). Default 3.
<code>spls_eta</code>	For "SPLS" only: sparsity grid passed to <code>spls::cv.spls()</code> .

Details

Method support. "PCR", "LASSO", and "SPLS" are fully adaptive and supported. "sPCR" is supported but requires `n_components` to be chosen by the caller (not cross-validated, matching the original script).

[FLAG] "SIR" is a simplified, *unvalidated* reimplementation using a single sliced-inverse-regression direction from `dr`, with no per-segment variable pre-selection. It intentionally does *not* reproduce the original script's SIR branch, which called an undocumented `edrSelec()` step from the non-CRAN package 'edr-GraphicalTools' that the original author flagged as never fully working ("NEED TO FIGURE OUT HOW TO INSTALL `edrGraphicalTools`!!"). `config.yml` in the original repository likewise documented SIR as "still under construction." Treat `method = "SIR"` output with corresponding caution.

[FLAG] The segmentation/calibration-window constraint (`chunk` must be small enough that even the shortest segment still spans all of `calib_years`) mirrors an explicit `stop()` check in the original script. I ported

that check explicitly rather than letting a mismatched-length input fail with a cryptic error deep inside `lm()` — but the underlying design constraint itself is inherited from the original code, not something I introduced.

Required packages: "LASSO" needs `glmnet`; "sPCR" needs `superpc`; "SPLS" needs `spls`; "SIR" needs `dr`. All are **Suggests**, only required for the method actually called.

Value

A list with **segments** (data.frame, one column per segment), **composite** (data.frame with columns **year** and **RP1**, the series to pass to `fit_baygmst()`), and **method**.

6 fit_baygmst

Fit the BayGMST Bayesian hierarchical AR(1) reconstruction model

Description

Prepares the data list expected by the `baygmst` Stan model and samples from it with `cmdstanr`, via `instantiate`. Ported from the data-preparation and `mod$sample()` block of the original `BayGMST_v1.0.R` script, restructured to take explicit arguments instead of reading `config.yml` and hardcoded relative paths, and to return an object rather than leaving variables in the global environment.

Usage

```
fit_baygmst(proxy, instrumental_T, forcing_G, forcing_V, forcing_S, years,
  chains = 4, parallel_chains = 2, iter_warmup = 500,
  iter_sampling = 1500, seed = NULL, ...)
```

Arguments

<code>proxy</code>	Representative proxy series – see <code>reduce_proxies()</code> . No missing values.
<code>instrumental_T</code>	Instrumental temperature, NA for pre-instrumental years to reconstruct.
<code>forcing_G</code> , <code>forcing_V</code> , <code>forcing_S</code>	Already-transformed forcings – see <code>transform_forcings()</code> . No missing values.
<code>years</code>	Calendar years, defining row order for all arguments above.
<code>chains</code> , <code>parallel_chains</code> , <code>iter_warmup</code> , <code>iter_sampling</code> , <code>seed</code>	Passed to <code>cmdstanr</code> 's <code>\$sample()</code> . <code>iter_sampling</code> $\geq$ 1000 required.
<code>...</code>	Additional arguments passed to <code>\$sample()</code> .

Value

An object of class "baygmst_fit": a list with `fit` (the `CmdStanMCMC` object), `years`, `idx_obs`, `idx_mis`, `data`, and `call`.

Note

[GAP] This function does not write any files by default (no `auto-ggsave()`, no auto-CSV export) – a deliberate deviation from the original script, which wrote directly to `cfg$folder_paths$...`. This matches CRAN policy on functions not writing to the filesystem by default, but it does mean any code that relied on those side-effect files (e.g. `outputs/reconstructions/fit_post_summaries.csv`) needs to be updated to call `write.csv()` itself on the returned object.

7 `cv_baygmst`

K-fold cross-validate the BayGMST model over the instrumental period

Description

Partitions the instrumental years of `instrumental_T` into `nfold` contiguous blocks; for each, refits the model with that block hidden, and reports recovery accuracy. Ported from the fold loop in `cv_v0.1.R`.

Usage

```
cv_baygmst(proxy, instrumental_T, forcing_G, forcing_V, forcing_S, years,
  nfold = 3, chains = 4, parallel_chains = 2, iter_warmup = 500,
  iter_sampling = 1500, seed = NULL, ...)
```

Arguments

Same as `fit_baygmst()`, plus `nfold` (number of folds, default 3).

Value

An object of class `"baygmst_cv"`: `folds` (combined data.frame of held-out predictions), `r2` and `mse` (per-fold, posterior mean of the Stan-computed `r2_cv/mse_cv` generated quantities), `years`, `call`.

Note

[FLAG] Two blocks in the original `cv_v0.1.R` were dropped, not ported: (1) a `Forcings.projections/RCP-scenario` block that was built but never referenced again (dead code), and which additionally hardcoded a path to a specific contributor's machine (`/Users/tylerbagwell/...`); and (2) a large commented "OLD" duplicate of the fold loop at the end of the file. Confirm nothing downstream still depended on either.

[FLAG] This implementation reports the fully Bayesian posterior mean of the Stan model's own `r2_cv/mse_cv` generated quantities. The original (non-dead) R loop instead computed a plug-in point-estimate R^2 in R from the posterior mean reconstruction. The two will be close but not identical – flagging the substitution explicitly rather than silently changing what "R2" means.

8 reconstruct

Extract a tidy reconstruction from a fitted BayGMST model

Description

Builds a tidy data.frame combining observed instrumental temperatures with the posterior reconstruction of pre-instrumental years – equivalent to the `gmst_reconstruction_data.csv` the original script wrote to disk as a side effect, but returned as an object instead.

Usage

```
reconstruct(fit, probs_inner = c(0.16, 0.84), probs_outer = c(0.025, 0.975))
```

Value

A data.frame with columns `year`, `type` ("instrumental"/"reconstruction"), `T_obs`, `T_mean`, `T_lo_inner`, `T_hi_inner`, `T_lo_outer`, `T_hi_outer`.

9 `plot_reconstruction`, `plot_trace`, `plot_posterior_densities`, `plot_cv`

Plotting functions

Description

- `plot_reconstruction(fit, title = "GMST Reconstruction")` – ported from the `p_ts` block of BayGMST_v1.0.R.
- `plot_trace(fit, parameters = ...)` – ported from the trace-plot block.
- `plot_posterior_densities(fit, parameters = ...)` – ported from the posterior-histogram block.
- `plot_cv(cv)` – ported from the `p_ts_cv` block of `cv_v0.1.R`.

All four return a `ggplot` object instead of calling `ggsave()`/auto-printing as a side effect; save with `ggplot2::ggsave()` if a file is wanted. **[FLAG]** The original script combined the reconstruction and posterior-density panels into one figure via `patchwork (p_ts + p_hist)`. Splitting these into separate functions means that combination is no longer automatic – users who want the original combined layout can do `plot_reconstruction(fit) | plot_posterior_densities(fit)` themselves via `patchwork` (moved to `Suggests`, since the package itself no longer calls it directly).

10 print/summary methods

S3 methods

Description

- `print.baygmst_fit(x, ...)` – one-line-per-field overview (year range, chains/draws, call).
- `summary.baygmst_fit(object, ...)` – posterior parameter summary (`alpha0`, `alpha1`, `phi_R`, `phi_T`, `beta0`, `betaG`, `betaS`, `betaV`, `sigma_y`, `sigma_z`) plus instrumental-period fit performance (RMSE, MAE, Bias, Correlation, R^2 , detrended R^2), replacing the `perf_stats/R2_dt` block of `BayGMST_v1.0.R`.
- `print.summary.baygmst_fit(x, ...)` – pretty-prints the above.
- `print.baygmst_cv(x, ...)` – per-fold and median R^2 .

Known limitations index

[FLAG] `src/stan/baygmst.stan`'s `y_ins_fitted` block (`t == 1` branch) indexes `y/z` with the count `NT_mis` rather than `idx_mis[NT_mis]`, flagged by the original author with `// CHECK THIS LINE`. On inspection (see §1 and `NEWS.md`), this is numerically correct given how this pipeline's inputs are always laid out (missing years contiguous at the start, observed years contiguous at the end) – *not* the live bug I initially reported it as – but it is fragile, undocumented shorthand that would break under non-contiguous missingness, and its effect is narrowly scoped to `y_ins_fitted[1]` alone, not to the parameter posteriors or to `y_mis`. Left unchanged, since even a belt-and-suspenders self-documenting rewrite is a code change to model machinery that deserves Tyler/Julien's sign-off, not a unilateral one.

[FLAG] Several `library()` calls in the original scripts (`car`, `fda`, `ggmap`, `maps`) had no evident corresponding usage in the code as written and were not carried into `DESCRIPTION`. If a hidden use is found, re-add them.

[GAP] `DESCRIPTION`'s `RoxygenNote` field is a placeholder (7.3.2) – `devtools::document()` will set the real value.

[GAP] `NAMESPACE` and every file in `man/` are hand-authored placeholders (see the header comment in each), not actual `roxygen2` output, because this restructuring was done without an R installation available to run it. `CRAN-READINESS.md` has the exact commands to regenerate and verify them for real.

[FLAG] references.bib entries were copied directly from Bagwell_thesis_proposal_for_Julien/ref.bib and JEG_biblio.bib – not reconstructed from memory.

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